Object Oriented Design and Implementation of a Military Aviation Spare Parts Inventory System

First Semester 2026-27
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SS G514 OBJECT ORIENTED ANALYSIS & DESIGN

Area: 40, Defence

STUDENT DETAILS	
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Course	SS G514 Object Oriented Analysis & Design
Paper selected	The efficiency of using a tailored inventory management system in the military aviation industry

Programming language and requirements

Item	Requirement
Language	Java 17 source compatibility; source compiles with JDK 17 or later.
Build	Apache Maven 3.8+; JUnit 5 dependency for tests.
Hardware	Any standard computer capable of running a Java 17+ JDK. No special hardware is needed.
Other software	Visual Studio Code with Extension Pack for Java is optional. No database server or third-party runtime package is required.
Data	Local CSV files are created in inventory-data/ when the program first runs.
Network	Maven may need network access the first time to download JUnit and build plugins.

Build, test, and run

1. Open the project folder in VS Code or a terminal.
2. Select JDK 17 or later for Maven. On macOS, set JAVA_HOME to the installed JDK before building.
3. Run mvn test. The submitted test run completed 6 tests with 0 failures.
4. Run java -cp target/classes h2025H1030207P_D2 from the project root.

Demo accounts and permissions

Login	Role	Main actions
keeper	Store keeper	Add/update parts, receive and issue stock, repair returns, reports.
tech	Technician	Raise requests, declare AOG, fit issued parts, reports.
planner	Planner	Select policy, review and approve proposals, approve non-critical requests.
commander	Commander	Approve critical-part requisitions, view reports and AOG.

Repeatable AOG demonstration

1. Log in as planner, choose menu option 10, select minmax, then log out. Set the policy before the request so its proposal uses the chosen strategy.

2. Log in as tech and choose option 2. Enter tail AC-01, priority aog, one line for ROT-220 quantity 1, and grounded y. The sample starts with no ROT-220 stock.

3. Log in as planner. Option 11 shows proposals; option 12 approves ROT-220.

4. Log in as keeper. Option 17 receives the approved proposal. Option 1 lists the generated serial.

5. Log in as commander. Option 3 approves the critical requisition. Log in as keeper and option 5 issues it.

6. Log in as tech. Use option 1 to find the serial, option 6 to fit it, option 14 for the AOG report, and option 13 as keeper for the stock report.

7. Choose option 19 to compare fixed-level and minimum-maximum policy results on synthetic demand.

Sample files

The submission bundle includes h2025H1030207P_D2.ip1 with one input response per console prompt and h2025H1030207P_D2.op1 with the captured output. All catalogue and aircraft data are fictional. To restore fresh initial state, close the program and remove its inventory-data folder.